

Education

- 2023-present **University of Waterloo**, Combinatorics and Optimization, Waterloo, ON, CANADA
PhD, Combinatorics and Optimization. GPA 88.0/100.0
- 2021-2023 **New York University**, Courant Institute of Mathematical Sciences, New York, NY, USA
Master, Mathematics. GPA 3.8/4.0
- 2017-2021 **University of Michigan Ann Arbor**, College of Literature, Science, and the Arts, Ann Arbor, MI, USA
Bachelor of Science, Honor Mathematics, Computer Science. GPA 3.7/4.0

Research/Internship Experience

- 2023-present **Critical Points at Infinity, University of Waterloo**. Supervised by Stephen Melczer
Using toric variety to explore proper compactification reflecting the properties of infinity points of algebraic varieties.
- 2023 Summer **TensorFlow Lite Performance Benchmarking, Google, Inc.** Supervised by Vamsi Manchala
Proposed and implemented benchmark algorithms comparing multiple on-device framework for high-performance ML deployment on edge platforms.
- 2022 Summer **Algebraic Geometry Solver, Google, Inc.** Supervised by Todor Milev, Michal Cierniak
Realized end to end algebraic geometry solver using Wu's Method, serving as a complementary method of rule-based solver for automatic geometry theorem proving.
- 2020 Summer **Efficient Eigensolvers and Applications, ICERM, Brown University** Supervised by Yanlai Chen
Implemented page rank algorithms and applied on website domains to obtain webpage rank scores; Developed performance comparison code frames; Built up web crawlers with directed graph generators.
- 2019 Summer **Natural Language Processing Optimizer, Facebook, Inc.**
Worked in Facebook Artificial Intelligence Applied Research Organization, Language and Translation Technologies team; Maintained and optimized PyText, an NLP modeling framework based on PyTorch; Implemented mixed precision training; Designed and optimized memory-efficient fp16 optimizers.

Publications & Presentations

- Toric compactifications for analytic combinatorics*, 2025, preprint.
- An exceptional equinumerosity of lattice paths and Young tableaux*, 2025, arXiv:2506.03116
- OGMC *Toric compactifications and critical points at infinity in analytic combinatorics*, Ontario Graduate Mathematics Conference, 2025
- Master Thesis *Optimizing surface closeness measurements: an analysis of distance field and adaptive simplification methods in tetrahedral meshing*, 2023.
- Undergraduate Thesis *Deformations of the origami hexagon, flipping operations, and pythagorean 4-tuples*, co-authored with Harry Richman, 2021, preprint.
- SIAM** *Origami hexagon deformations and flip graph*, SIAM Annual Meeting, 2021.
- SPIE** *U-Net-based deep learning for detection and segmentation of vertebral bodies in spinal MRI scans*. SPIE Medical Imaging, Huston, TA, 2020.

Honors & Fellowships

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| 2025 | Equity, Diversity, and Inclusion (EDI) Award | 2021 | SIAM Annual Meeting Student Travel Award |
| 2024-2025 | Women in Mathematics Mentorship Award | 2020 | Summer Undergraduate Research Fellowship |
| 2022 | Summer@ICERM Reunion Funding | 2017-2020 | University Honors |

Technical Skills

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| Programming | C++, PYTHON, JAVA, JULIA, SAGEMATH | Machine Learning | PYTORCH, PYTEXT, TENSORFLOW |
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